Solution of Rational Functions

Course: Pre-Calculus-I

AH Sheikh

26CS(Evening)- Fall 2026

Outline

1 Solving Rational Equations

2 Solving Rational Inequalities

3 Applications and Graphs

4 Class Practice

What Does "Solve" Mean?

Definition

To solve a rational equation

$$\frac{P(x)}{Q(x)} = \frac{R(x)}{S(x)}$$

means to find all real x in the domain for which the two sides are equal.

Key idea: clear denominators by multiplying both sides by the least common denominator (LCD).

Caution: Multiplying by an expression that can be zero may introduce extraneous solutions. Always check candidates in the *original* equation.

Note

A solution that makes any original denominator zero is not valid.

Method: Solving Rational Equations

1. Find the domain (exclude values making any denominator zero).
2. Determine the LCD of all fractions.
3. Multiply every term by the LCD; simplify.
4. Solve the resulting polynomial equation.
5. Check each candidate against the domain; discard extraneous roots.

Special case: one fraction equals a polynomial

$\frac{P(x)}{Q(x)} = 0 \iff P(x) = 0$ **and** $Q(x) \neq 0$. This is the equation behind the roots of a rational function.

Worked Example 1: Simple Linear Denominators

Solve: $\dfrac{2}{x} + \dfrac{3}{x+1} = 1.$

Domain: $x \neq 0,\ x \neq -1.$ **LCD:** $x(x+1).$

Multiply through:

$$2(x+1) + 3x = x(x+1).$$

Simplify:

$$2x + 2 + 3x = x^2 + x \implies x^2 - 4x - 2 = 0.$$

Quadratic formula:

$$x = \frac{4 \pm \sqrt{16+8}}{2} = \frac{4 \pm 2\sqrt{6}}{2} = 2 \pm \sqrt{6}.$$

Both $2+\sqrt{6}$ and $2-\sqrt{6}$ are in the domain. ✓

Worked Example 2: Extraneous Root

Solve: $\dfrac{x}{x-2} = \dfrac{2}{x-2} + 3.$

Domain: $x \neq 2$. **LCD:** $x - 2$.
Multiply through:

$$x = 2 + 3(x-2) = 2 + 3x - 6 = 3x - 4.$$

Solve:

$$x = 3x - 4 \implies -2x = -4 \implies x = 2.$$

But $x = 2$ is excluded by the domain! So $x = 2$ is an extraneous solution.

Note

Conclusion: the equation has **no solution** *over* $\mathbb{R}$.

Worked Example 3: Quadratic Denominator

Solve: $\dfrac{1}{x-1} - \dfrac{1}{x+1} = \dfrac{2}{x^2-1}$.

Note: $x^2 - 1 = (x-1)(x+1)$. Domain: $x \neq \pm 1$.
LCD: $(x-1)(x+1)$. Multiply through:

$$(x+1) - (x-1) = 2.$$

LHS simplifies to 2:

$$2 = 2.$$

This is an identity on the domain.

Solution: all real x with $x \neq \pm 1$, i.e. $x \in \mathbb{R} \setminus \{-1, 1\}$.

Solving Rational Inequalities: Critical Points

Method

To solve an inequality such as $\frac{P(x)}{Q(x)} > 0$:

1. Factor numerator and denominator.
2. Find critical points: zeros of P (numerator) and zeros of Q (vertical asymptotes).
3. Plot them on a number line; these split $\mathbb{R}$ into intervals.
4. Test one point per interval in the original inequality.
5. Include endpoints only if the inequality is non-strict and the endpoint is not a denominator zero.

Note

Never multiply across by a denominator whose sign you don't know — use the sign-chart method instead.

Worked Example: Rational Inequality

Solve: $\dfrac{x-1}{x+2} \geq 0.$

Critical points: $x = 1$ (numerator zero), $x = -2$ (denominator zero).

Test intervals:

Interval	Test x	Sign of $\frac{x-1}{x+2}$	Include?
$(-\infty, -2)$	-3	$(-)/(-) = +$	yes
$(-2, 1)$	0	$(-)/(+) = -$	no
$(1, \infty)$	2	$(+)/(+) = +$	yes

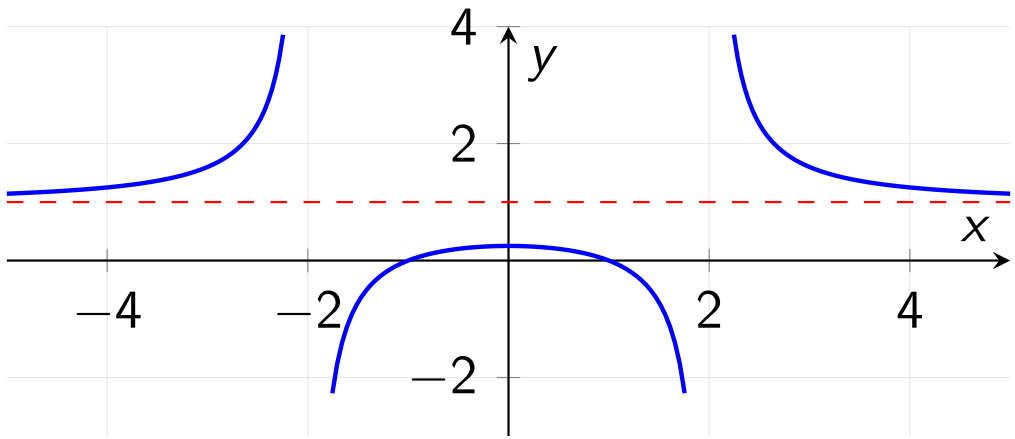
Endpoint $x = 1$ is included (numerator zero, denominator nonzero); $x = -2$ is excluded (vertical asymptote).

$$\boxed{x \in (-\infty, -2) \cup [1, \infty)}$$

Graph of a Rational Function: Full Analysis

Analyse and sketch $f(x) = \dfrac{x^2 - 1}{x^2 - 4}$.

- **Domain:** $x \neq \pm 2$.
- **Vertical asymptotes:** $x = -2$ and $x = 2$.
- **Horizontal asymptote:** equal degrees, leading coeffs $1, 1 \Rightarrow y = 1$.
- x**-intercepts:** $x^2 - 1 = 0 \Rightarrow x = \pm 1$.
- y**-intercept:** $f(0) = 1/4$.
- **Symmetry:** $f(-x) = f(x) \Rightarrow$ even (symmetric about y-axis).



Solving a Rational Equation Graphically

To solve $f(x) = g(x)$ where both are rational:

- Plot $y = f(x)$ and $y = g(x)$.
- The solutions are the x-coordinates of intersection points.
- Any intersection at a vertical asymptote of either function is spurious.

Example: Solve $\dfrac{1}{x} = x - 1$.

$$1 = x(x-1) = x^2 - x \implies x^2 - x - 1 = 0,$$

$$x = \frac{1 \pm \sqrt{5}}{2}.$$

These are the golden-ratio points φ and $-\varphi^{-1}$.

Note

Graphical solutions should always be verified algebraically when possible.

Worked Application: Work-Rate Problem

Problem: Pipe A fills a tank in 4 hours; Pipe B fills it in 6 hours. How long will both take together?

Setup: Let t = time (hours) working together.

$$\frac{1}{4} + \frac{1}{6} = \frac{1}{t}.$$

LCD: $12t$. Multiply through:

$$3t + 2t = 12 \implies 5t = 12 \implies t = \frac{12}{5} = 2.4 \text{ hours}.$$

Check: In 2.4 h, A fills $2.4/4 = 0.6$ of the tank, B fills $2.4/6 = 0.4$, total 1.0. ✓

Note

Work-rate problems always reduce to an equation of the form $\frac{1}{a} + \frac{1}{b} = \frac{1}{t}$.

Class Practice Questions

P1. Solve $\dfrac{3}{x-2} = \dfrac{5}{x+1}$.

P2. Solve $\dfrac{x}{x-1} - \dfrac{2}{x} = 1$.

P3. Solve $\dfrac{x^2-4}{x-1} = 0$ and state any excluded values.

P4. Solve the inequality $\dfrac{x+3}{x-2} < 0$.

P5. A tank can be filled by Pipe A in 5 hours and drained by Pipe B in 8 hours. If both are open, how long until the tank is full?

P6. Sketch $f(x) = \dfrac{x+2}{x^2-1}$: find domain, asymptotes, intercepts, and sign chart.

Note

For P1–P4, always state the domain before clearing denominators, and check for extraneous roots.